MOVIE RENTAL DATA ANALYSIS

Final Analytical Report | MECE framework

This report consolidates the supplied SQL question workbook, Power BI dashboard PDF, and Power BI solved-question PDF into a structured business review. Each analysis question is assigned one primary workstream; source inconsistencies are documented rather than silently reconciled.

$67,406.56	16,044	599	$112.53
Total sales revenue	Rental transactions	Active customers	Average spend/customer

Business question

How can the rental business understand revenue contribution, customer value, content demand, and operational throughput—and use those findings to prioritize actions?

Executive summary

The outputs report $67,406.56 in sales revenue, 16,044 rental transactions, and 599 active customers. The SQL customer segments show 349 Loyal High-Frequency customers contributing $44,297.76 (about 65.7% of reported revenue), versus 250 Occasional Repeat customers contributing $23,108.80. Revenue is concentrated among a small group of countries; the ten most-demanded titles shown have 31–34 rentals each. These findings support retention and availability reviews, but period and metric inconsistencies should be validated before financial reporting.

Use caveat: This is a synthesis of visible outputs. SQL was not rerun against the original database, and ambiguous chart labels or conflicting figures are not treated as resolved facts.

Sources reviewed

Movie_Rental SQL_Excel_file.xlsx; movie rental Dashboard.pdf (3 pages); movie rental powerbi solved question.pdf (15 pages).

1. Scope, Methodology & MECE Structure

MECE means Mutually Exclusive, Collectively Exhaustive. Each question has one primary analytical home; the combined workstreams cover the main business questions in this project.

Primary workstream	Questions owned	Business use
A. Revenue & financial performance	Total sales; monthly revenue; country revenue; store contribution.	Understand scale, trend, geography and store contribution.
B. Customer & market analysis	Active customers; average spend; segments; VIPs; frequency and customer geography.	Identify valuable groups and retention opportunities.
C. Inventory & content performance	Inventory by rating/category; title demand; category rental-rate comparisons; allowed rental duration.	Guide availability, merchandising and pricing review.
D. Operational performance	Rental volume; staff throughput; actual duration where defined; metric validation.	Review workload and operational data quality.

Method

Step	Approach
1. Consolidate	Read the SQL workbook outputs and supplied dashboard/solved-question PDFs.
2. Select evidence	Prefer explicitly labeled values; distinguish exact workbook figures from rounded dashboard cards.
3. Assign MECE ownership	Place each question under revenue, customer, content, or operations.
4. Interpret carefully	Separate observed values from implications and recommendations.
5. Quality review	Record mismatched periods, labels and counts instead of guessing which is correct.

Definitions and limits

“Rental transactions” means rental records reported in the outputs, not unique customers. “Allowed rental duration” is a permitted period and is not actual rental duration. Category rental-rate values are reproduced without assigning a currency because the chart does not explicitly label one. This report does not independently recalculate results from raw rows.

Coverage: The report summarizes visible figures; it does not claim to reproduce every solved question or independently validate the underlying database.

2. Revenue & Financial Performance

2.1 Revenue scale and trend

The SQL workbook reports total sales revenue of **$67,406.56**; the dashboard rounds this to **$67.4K**. The SQL rental count is 16,044, consistent with the dashboard's rounded 16K headline.

Period in SQL output	Revenue
May 2005	$4,823.44
June 2005	$9,629.89
July 2005	$28,368.91
August 2005	$24,070.14
February 2006	$514.18

July 2005 is the highest listed month ($28,368.91), followed by August ($24,070.14). February 2006 is much smaller. The outputs do not establish whether the month list is complete, filtered, or drawn from different reporting windows; do not infer seasonality from it.

2.2 Country revenue contribution

Country	Revenue	Customers shown
India	$6,628.28	60
China	$5,798.74	53
United States	$4,110.32	36
Japan	$3,470.75	31
Mexico	$3,307.04	30
Top-five combined	$23,315.13	210

The top five listed countries contribute $23,315.13, approximately 34.6% of total revenue. India is the largest listed contribution at $6,628.28. This supports further market-level analysis but does not establish why these countries perform better.

2.3 Store sales

The dashboard visual shows Store 2 at 33,730 and Store 1 at 33,269. The measure/unit is not fully explicit in the extracted label, and related SQL outputs do not consistently align. Treat these as dashboard-displayed values pending validation.

Revenue takeaway: Validate monthly completeness and store metric definitions before making trend or store-performance decisions.

3. Customer & Market Analysis

3.1 SQL customer segments

Segment	Customers	Revenue	Average spend/customer
Loyal High-Frequency	349	$44,297.76	$126.93
Occasional Repeat	250	$23,108.80	$92.44
Total	599	$67,406.56	—

Loyal High-Frequency customers represent 349 of 599 customers (58.3%) and contribute about 65.7% of revenue. Their average spend is approximately $34.49 higher per customer than the Occasional Repeat segment, making retention of this group a priority.

3.2 Dashboard frequency split

Frequency category	Customers	Share shown
High Frequency	409	68.28%
Medium Frequency	134	22.37%
Low Frequency	56	9.35%
Total	599	100%

This dashboard classification is distinct from the SQL workbook's two segments. Their rules are not supplied, so the two schemes should not be treated as interchangeable.

3.3 Top VIP values as displayed

Name	Value shown
Karl Seal	222
Eleanor Hunt	217
Clara Shaw	196
Marion Snyder	195
Rhonda Kennedy	195
Tommy Collazo	187
Wesley Bull	178
Tim Cary	176
Marcia Dean	176
Ana Bradley	175

The visual is labeled as top VIP spending customers, but the unit and exact measure are not sufficiently explicit; values are reproduced without assuming they are dollar amounts.

Customer takeaway: Protect and grow the Loyal High-Frequency segment; test reactivation for Occasional Repeat customers. Document segment rules and the VIP metric first.

4. Inventory & Content Performance

4.1 Inventory by film rating

Rating	Inventory count shown
PG-13	1,018
NC-17	944
PG	924
R	904
G	791

PG-13 is the largest listed rating group (1,018), followed by NC-17 (944). These are inventory counts, not rental demand or profitability; compare with title-level rental and availability data before changing stock.

4.2 Most-demanded titles

Title	Rentals shown
Bucket Brotherhood	34
Rocketeer Mother	33
Forward Temple	32
Grit Clockwork	32
Juggler Hardly	32
Ridgemont Submarine	32
Scalawag Duck	32
Apache Divine	31
Goodfellas Salute	31
Hobbit Alien	31

The top-ten list is tightly clustered at 31–34 rentals. Check stock availability and copies on hand before inferring that any title is understocked; the visual does not provide availability or time-window detail.

4.3 Category rental-rate comparison

Higher category	Rate shown	Lower category	Rate shown
Games	3.25	Action	2.65
Travel	3.24	Documentary	2.67
Sci-Fi	3.22	Classics	2.74
Comedy	3.16	Family	2.76
Sports	3.13	Animation	2.81

Category averages range from 2.65 to 3.25; the unit/currency is not explicit, so these values should not be interpreted as profit margins. Allowed rental duration ranges from 4.7 days (New and Sports) to 5.4 days (Travel), which is policy duration—not actual duration.

Content takeaway: Combine title demand with stock availability, inventory age and cost context before stocking or pricing changes.

5. Operations, Recommendations & Conclusion

5.1 Operational evidence and validation checks

Metric / visual	Values shown	Issue / interpretation
SQL staff rental counts	Mike Hillyer 8,054; Jon Stephens 7,990	Totals 16,044; consistent with SQL rental count.
Solved-question staff chart	Mike 8,040; Jon 8,004	Same total, but individual counts differ from SQL output.
Staff sales visual	Mike 33.92K; Jon 33.48K	Measure differs or is unclear versus rental-count visual.
Monthly revenue	May–Aug 2005 plus Feb 2006	Period may be incomplete or differently filtered.
Question/chart mismatch	Question says payment method; chart says sales by staff clerk	Question and visual do not match.
Question/chart mismatch	Question says employment duration; chart shows rental processing volume	Not evidence about employment tenure.
Question/chart mismatch	Question says customer ratings; chart shows country sales	Treat visual as country sales, not ratings.

5.2 Recommended actions

Priority	Action	Success measure
1. Data quality	Reconcile date filters, staff counts, store labels and question/chart titles.	Approved metric dictionary; dashboard cards agree with SQL outputs.
2. Customer retention	Test retention for Loyal High-Frequency customers and reactivation for Occasional Repeat customers.	Compare repeat-rental rate and revenue/customer with a control group.
3. Market focus	Review acquisition and repeat behavior in India, China, US, Japan and Mexico.	Incremental revenue and repeat rate by country.
4. Content availability	Audit stock availability for the ten most-demanded titles.	Out-of-stock events, rentals per available copy, and lost-demand proxies.
5. Operations	After validating staff metrics, compare workload and service outcomes by clerk.	Transaction counts plus service quality and timing—not counts alone.

Conclusion

The project outputs provide a useful first view of revenue, customer value, content demand and rental operations. The clearest commercial signal is the Loyal High-Frequency segment's contribution, alongside revenue concentration in the five highest listed countries. Content charts can guide follow-up checks but should not drive stock or pricing changes without availability and cost context. Resolve the listed data-definition and period mismatches, then refresh the report from a validated dataset.

Limitations

This report summarizes visible Excel/PDF outputs; it does not rerun SQL or independently calculate results from raw transaction rows. Some chart labels, units and periods are ambiguous. Conflicting values are documented rather than reconciled by assumption; recommendations are proposed actions, not proven outcomes.